Data Confidentiality or Network Supervision? Blockchain-Secured Hybrid LiFi/WiFi Architecture

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***Abstract*—Data confidentiality and network supervision are both necessary for secure communication systems against open-air transmissions' inherent vulnerability and openness. However, these two fundamental objectives often contradict each other. In this study, we propose a blockchain-secured hybrid LiFi/WiFi (B-HLW) architecture to resolve the dilemma between data confidentiality and network supervision while providing high-speed indoor data transmissions. Specifically, in the B-HLW, we separate the data plane from the control plane, and guarantee the data confidentiality in the data plane and supervise the behavior in the control plane. We present the B-HLW with a detailed design and illustrate the workflow through a specific case study based on Ethereum Testnet. We provide extensive simulation results to demonstrate the efficiency and scalability of the B-HLW. The source code of both the simulation and Ethereum Testnet is released at https://github.com/hjx983361377/B-HLW.**

***Index Terms*—Blockchain, data confidentiality, LiFi, network supervision, wireless security.**

I. Introduction

Wireless communications have revolutionized connectivity with their ease and efficiency, and served as the backbone for diverse sectors such as finance, industry, and even the military, but have also introduced serious security risks. Unlike wired links, wireless channels are easier to intercept and eavesdrop [1]. Ensuring the confidentiality and integrity of sensitive data in such open-air transmissions is therefore essential. At the same time, the openness that makes wireless systems flexible also increases their attack surface: adversaries can exploit wireless-specific vulnerabilities to cause data breaches, intrusions, or service disruption [2]. Effective network supervision thus requires monitoring individual nodes and information flows, via, for example, real-time behavioral analysis, anomaly detection, and traffic collection, while avoiding exposure of sensitive payloads. In a nutshell, both data confidentiality and network supervision must be jointly considered for secure wireless communications [3].

However, these two fundamental objectives–data confidentiality and supervision–are in conflict with each other. On the one hand, strong confidentiality safeguards payloads from interception and eavesdropping; however, it also hinders the ability to monitor network operations and control information. On the other hand, effective supervision often depends on traffic visibility, packet inspection, or flow analysis, which can conflict with privacy concerns. Consequently, this creates a practical dilemma: how can a wireless system preserve end-to-end confidentiality and still enable reliable, tamper-resistant supervision? Achieving both goals simultaneously is difficult, and the challenge is further compounded by the requirement that solutions must not sacrifice key performance indicators like throughput as an expense.

Though a variety of technologies have been developed to address either data confidentiality or network supervision, many existing solutions address one side of the problem but not both. Cryptography methods [4], artificial noise schemes [5], and security-oriented beamforming [6], to name just a few, can protect the transmitted data from eavesdroppers; however, they also reduce the information available for network supervision. Meanwhile, anomaly detection, auditing, and logging can detect misbehavior and network attacks by recording user activities and network events in audit logs or analyzing collected data [2]. Nevertheless, the collecting and recording of data by these technologies may increase the risk of data breaches and thus weaken data confidentiality.

Most recently, Liu *et al.* proposed a zero-trust-based 6G security architecture to achieve both goals of data confidentiality and network supervision [3]. This architecture employed physical-layer security techniques and lightweight attribute-based encryption to ensure data confidentiality. Meanwhile, data and behavior supervision of the service network (SN) and home network (HN) were performed by recording their actions on the blockchain. However, such supervision overlooked potential user misbehavior and still recorded certain critical data, which may lead to potential leaks. Ling *et al.* proposed the blockchain-based network hawkeye function (BNHF) by watching network entities and storing the records on the blockchain, as a digital forensics model, to simultaneously supervise network traffic and ensure data confidentiality and integrity [7]. With the pre-defined smart contract, the BNHF detected the mismatch between incoming and outgoing traffic by comparing their hash abstractions, thus preventing potential data breaches and tampering. Nonetheless, this work did not consider the supervision of the control plane, and the transmitted data are still visible in the physical layer. Keshk *et al.* proposed a privacy-preserving anomaly detection framework to prevent data leakage during network supervision [8]. This framework protected the sensitive data from exposure via some proper pre-processing, including mapping, reducing, and normalizing data features, and identified user misbehavior

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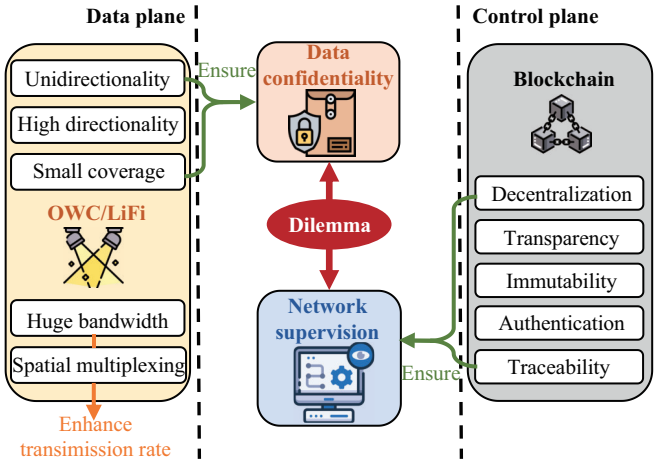


Figure 1: Principles of the B-HLW to resolve the dilemma between data confidentiality and network supervision. The properties of OWC and blockchain are utilized to protect payload data confidentiality in data plane and achieve network supervision in control plane.

and network intrusions by analyzing the pre-processed data instead of the sensitive raw data. Similarly, cryptography was also employed for the pre-processing and transmission of confidential data and combined with several supervision technologies in recent efforts [9]. While these pre-processing methods limit raw data exposure, they inevitably compromise the effectiveness of network supervision, such as reducing the accuracy of anomaly detection.

Note that optical wireless communication (OWC) has shown great potential in protecting data confidentiality [10]. Unlike traditional radio frequency (RF) communications, the optical communication medium, i.e., the light beam, is unable to propagate through opaque objects such as walls, and is highly directional as seen in extreme cases such as laser beams. Such a property of spatial limitation allows OWC to effectively safeguard data confidentiality while providing directional high-speed transmissions with huge free-to-use bandwidth [10]. Meanwhile, the one-way property of OWC can mitigate potential threats to data integrity and confidentiality posed by adversarial nodes belonging to the current network.

OWC is still incapable of monitoring the network and information flows therein. Conversely, its highly directional and short-range characteristics prevent it from observing information flows [10]. In this context, blockchain presents an innovative solution for decentralized and transparent supervision of individual nodes' behavior [11]. As a distributed ledger, blockchain allows data to be stored and operated in a decentralized manner while maintaining high integrity, credibility, and security. The transparency and immutability of on-chain data can easily trace and record every individual node's behavior [7], [12]. Additionally, the decentralized nature of blockchain can reduce security threats in existing wireless systems brought by their centralized architectures.

Therefore, we propose a blockchain-secured hybrid LiFi/WiFi (B-HLW) architecture by introducing OWC and blockchain into wireless systems to address the dilemma between data confidentiality and network supervision. As illustrated in Fig. 1, B-HLW separates the data and control planes: the data plane uses unidirectional OWC for payload delivery with strong physical-layer confidentiality, while the control plane uses blockchain to monitor and trace real-time network actions without compromising end-to-end privacy. By decoupling content delivery from supervisory functions, B-HLW enables tamper-resistant, decentralized supervision and preserves the confidentiality of transmitted payloads. Compared to existing works [8], [9], the B-HLW architecture does not record, collect, or analyze any specific payload data, nor does it employ any complex techniques that could degrade system performance. Furthermore, unlike conceptual frameworks, we implement the B-HLW based on Ethereum Testnet. (See the source code provided in the abstract.) Comprehensive simulations illustrate that the B-HLW can achieve a throughput higher than the Wi-Fi 6 [13] while ensuring confidentiality and supervision. To the best of our knowledge, this is the first work that leverages both the complementary advantages of OWC and blockchain to establish a new secure paradigm for wireless communications.

The main contributions of this work are summarized as follows:

- We propose the B-HLW, a hybrid LiFi/WiFi architecture secured by blockchain that separates the data and control planes. This decoupling preserves payload confidentiality and enables decentralized supervision.
- Practically, we design a dedicated blockchain for the B-HLW and also implement smart contracts to support secure transmission services under tamper-evident supervision.
- We present a proof-of-concept implementation on Ethereum Testnet to illustrate the effectiveness and feasibility of our proposed B-HLW architecture.
- We validate the proposed B-HLW's performance with a series of practical simulations and demonstrate the improvement in throughput, efficiency, and scalability with strong confidentiality and effective supervision.

The remainder of the paper is organized as follows. Section II presents the proposed B-HLW architecture. In Section III, we provide an Ethereum-based case study. Section IV shows our simulation results. Section V concludes the paper and discusses practical deployment challenges.

II. Blockchain-Secured Hybrid LiFi/WiFi

A. Scenario

We consider a hybrid LiFi/WiFi system in an indoor scenario (see Fig. 2). A WiFi AP is placed at the center of the room, and multiple laser diodes (LDs) are distributed on the ceiling as LiFi APs [10]. All LDs connect to a secure database that stores large-scale confidential data such as deep learning models, high-resolution videos, very large datasets, or real-time, precise maps. In this scenario, users may request specified data from the database on demand and expect to receive them securely and quickly through wireless transmissions. The data involved could be highly confidential and sensitive, requiring protection against unauthorized access,

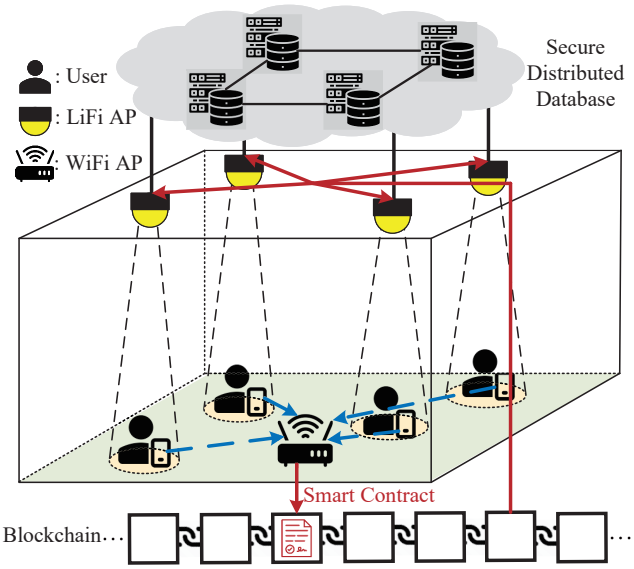


Figure 2: Illustration of blockchain-secured hybrid LiFi/WiFi architecture. WiFi is utilized for transmitting control messages, and OWC serves as the downlink for transmitting payload data from a secure distributed database. Blockchain acts as a public ledger to record users' digital activities and enforce digital actions in smart contracts.

interception, and tampering during the transmissions. Meanwhile, the system must support comprehensive supervision of network operations and user behavior to promptly detect, trace, and mitigate malicious activities. Therefore, both strict data confidentiality and effective monitoring must be ensured simultaneously.

The above scenario is common across many sectors of modern society. The users in the B-HLW could be either human beings or machines. For example, in the financial sector, human traders and automated trading agents may request proprietary assets such as market models, strategy parameters, transaction data, and execution logs. These data are commercially valuable and sensitive to clients' privacy, and are often subject to strict regulatory and contractual obligations. Protecting them therefore requires strong payload confidentiality to prevent model theft, reverse engineering, or the interception that could enable market manipulation. At the same time, the financial system requires real-time supervision to enforce compliance rules, detect anomalous information flows, and support rapid forensic analysis [1]. For another instance of industrial automation, users are primarily machines such as controllers, robotic manipulators, and edge servers, and will continuously request operational data such as trained predictive models, sensor measurements, or high-fidelity digital twins. These proprietary data assets must remain confidential to protect intellectual property. Continuous supervision is necessary to detect anomalous behavior and enforce fine-grained role-based access control. The injection into the database through the open-air link must be prevented to avoid unauthorized command insertion or corrupted firmware delivery [7].

Therefore, the above examples present how the core dilemma arises from maintaining data confidentiality and ensuring network supervision. Apparently, it is challenging to satisfy users' demands for both objectives. To address the dilemma without sacrificing the transmission rate, we present the B-HLW architecture with a detailed design and workflow in the following subsections.

B. Overview

Now we would like to show the big picture of the proposed B-HLW scheme. Fig. 2 shows the fundamental architecture of B-HLW and how the WiFi AP and LDs interact with the underlying blockchain. In the B-HLW, the blockchain acts as a public ledger to record digital activities by users and enforce digital actions in smart contracts. A majority of miners are assumed to be honest and reliable in maintaining a benign blockchain. In the B-HLW, WiFi links are utilized for transmitting control messages, such as user requests and acknowledgments (ACKs), and OWC links serve as downlinks for transmitting payload data. (In certain scenarios with high security requirements, control messages could be further encrypted with public-key cryptography (e.g., Rivest-Shamir-Adleman (RSA) algorithm) and decrypted only by authorized parties to reduce the risk of data breaches, which, however, will compromise the effectiveness of network supervision.) The confidential data is only transmitted unidirectionally from the database to users, and users are unable to modify the data in the database or inject anything into the database thanks to the one-way property of OWC links. In the following, we will present the design of the B-HLW blockchain and smart contracts, which are used to handle user requests, supervise user behavior, and ensure appropriate downlink transmission scheduling.

C. Blockchain and Smart Contract Design

In this subsection, we show the B-HLW design regarding smart contract and blockchain (see Fig. 3(a)). We summarize the data fields in the B-HLW blockchain, such as parent hash, timestamp, Merkle Root, etc., and explain their meanings. Specifically, we add a new data field, the service list, into the block body of the B-HLW dedicated blockchain. The service list is a table whose each column corresponds to an LD and whose entries record the user's identifiers (IDs) that the LD is responsible for serving. To reduce the storage overhead and support efficient updates, the service list in the newly-generated block includes only the newly-added requests. The blockchain thus maintains a global service list incrementally by recording the service list from the genesis block to the latest one. Note that, to limit user waiting time, each LD column can only accommodate a given number of unfinished requests (three in Fig. 3(a)).

Now, we show how we design the secure transmission smart contract (STSC) for establishing and enforcing secure transmissions in the B-HLW. We implement the STSC as on-chain script code and include the following functions: `initializeServiceList()`, `updateServiceList()`, `constructor()`, and `findPair()`. (See Fig. 3(a) for detailed descriptions.) The function `constructor()` is used to deploy the

Block Structure

Block Height / Number		
Part	Field	Explanation
Block Head	Parent Hash	Hash of the previous block
	Timestamp	Time when the block was created
	Merkle Root	Root of the Merkle Tree
	Miner Address	Address of the miner that generated the block
	Nonce	A random value that satisfy the consensus problem
Block Body	Request Transactions	Transaction information of users requesting service and calling the smart contract
	Service List	A table to record order of users served by LDs

Service List Structure

Order	LD 1	LD 2	···	LD N
1	User 6	User 3	···	User 2
2	User 15	User 8	···	
3		User 11	···	

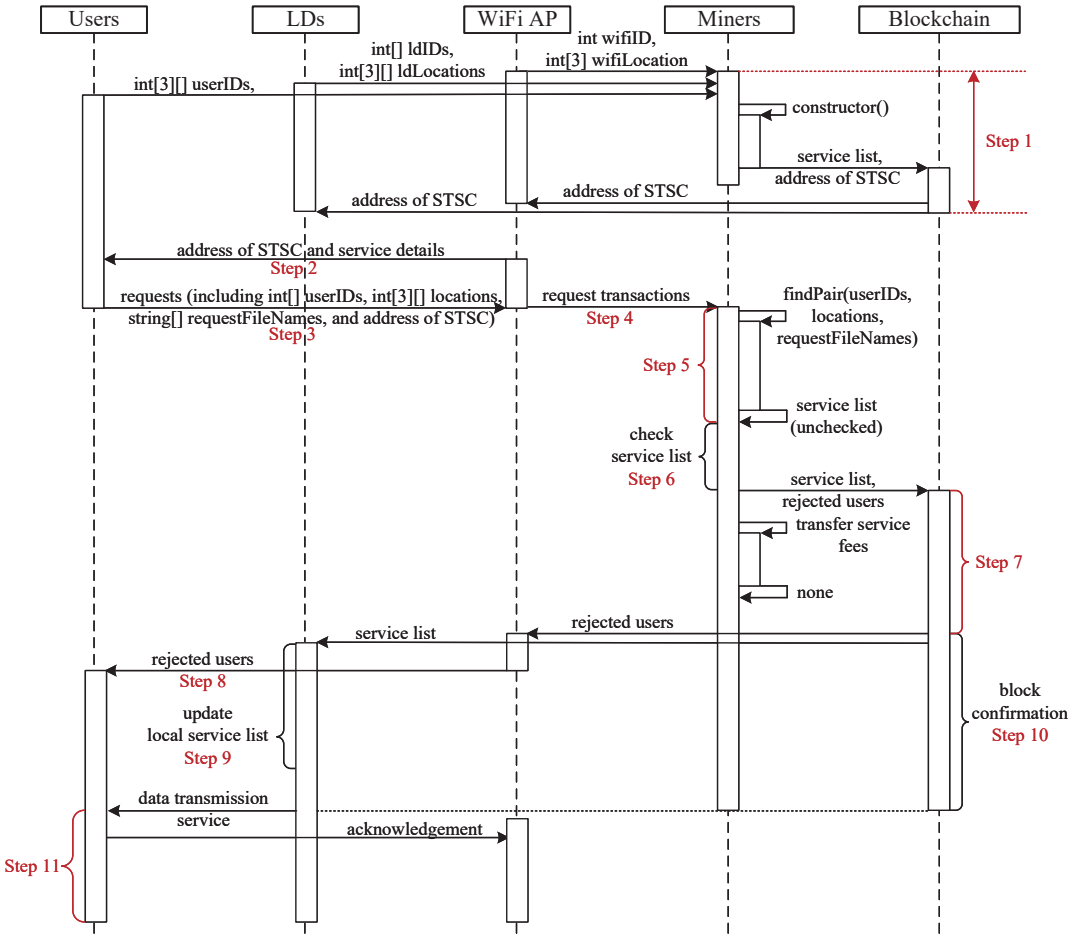
Secure Transmission Smart Contract (STSC)

Address			
Function	Input	Output*	Explanation
constructor()	int[] userIDs, int wifiID, int[3] wifiLocation int[] ldIDs, int[3][] ldLocations	Service list (initialized), smart contract address	Deploy the smart contract and execute initializeService List()
initializeServiceList()	None	Service list (initialized)	Initialize the service list
findPair()	int[] userIDs, int[3][] locations, string[] requestFileNames	Service list (unchecked)	Find appropriate LD for each user and execute updateService List()
updateServiceList()	int[] userIDs, int[] pairedLdIDs	Service list (unchecked)	Update the service list based on pairing results

update

*Output: Denotes the modification of the smart contract state or block data rather than the actual function output. All functions have no actual output.

(a)



(b)

Figure 3: Illustration of the B-HLW design and workflow. (a) Data structure of the block, smart contract, and service list in the B-HLW and their relationships. The top-left panel displays the data fields in the dedicated B-HLW blockchain (such as parent hash, timestamp, Merkle root, the service list, etc.). The service list is a table whose each column corresponds to an LD and whose entries record the user IDs that the LD is responsible for serving. The smart contract is implemented as on-chain scripts and includes functions for deployment and downlink scheduling. (b) Message interaction diagram of the workflow. In Step 1, necessary IDs and locations from users, WiFi AP, and LDs are transmitted to the miner network for contract deployment on the blockchain. Steps 2 to 10 illustrate the message flow of a request, from its upload to be processed, among different entities. In Step 11, users receive the requested data from LDs and feedback through the WiFi AP.

STSC on the blockchain. Once the STSC's deployment transaction, which is signed by the WiFi AP, LDs, and users to guarantee the integrity and mutual agreement, is included in a main-chain block, the function `constructor()` is called to create an STSC instance with its address recorded in the very block. As this block is accepted by the main chain, the STSC is successfully deployed on-chain, and authorized parties can obtain this instance's on-chain address and its application binary interface (ABI) to call it. The function `constructor()` also invokes the function `initializeServiceList()` to initialize the service list field on the blockchain. The function `findPair()` is the core of the B-HLW and enables effective downlink transmission scheduling for users who upload requests. It assigns the most appropriate LD for each user based on the information in its request. The matching policy in the function `findPair()` is pluggable and may incorporate, e.g., link strength, request priority, LD load, or content locality. The function `updateServiceList()` is called by the function `findPair()` to update the service list in the new block based on assignment results. All authoritative actions (deployment, assignments, and updates) require digital signatures from the relevant parties to ensure tamper-evident authorization and non-repudiation.

The dedicated blockchain and STSC form the foundation for network supervision in the control plane. User requests are automatically processed by the pre-defined STSC, and the assignment results are recorded in the service list field. The transparency and immutability of on-chain data ensure that user behavior can be effectively audited and supervised. Note that the entire request processing occurs only in the control plane and does not involve confidential data. LDs do not directly handle user requests; instead, they only know the IDs of the users to be served through the on-chain service list and then transmit the corresponding data. This design ensures that any operations or supervision in the control plane will not compromise the confidentiality of the data plane.

D. Workflow

In this subsection, we show how the B-HLW operates to achieve both data confidentiality and network supervision. The detailed steps of the workflow are presented as follows, with the interactions between multiple components illustrated in Fig. 3(b).

- **Step 1: Contract Deployment**. The WiFi AP, LDs, and users sign the agreed STSC and submit it to the blockchain network via a deployment transaction. Miners deploy the STSC by executing the function `constructor()` to create an instance of the STSC on-chain.
- **Step 2: Service Availability Broadcast**. The WiFi AP periodically broadcasts the address of the STSC instance and service details (such as fees and achievable transmission rates) to all users to announce the service availability.
- **Step 3: Request Upload**. Users who want to download specified data from the database upload requests to the WiFi AP. The request includes the user's ID and location, the requested data file name, and the STSC's address.
- **Step 4: Transaction Submission**. For each request, the WiFi AP constructs a transaction from its own address to the STSC's address. This transaction includes an input data field that calls the function `findPair()` of the STSC and attaches the user's ID, location, and requested data file name as the parameters. Then the WiFi AP submits it to the blockchain network.
- **Step 5: Contract Execution**. After validating all transactions, miners execute the function `findPair()` to allocate LDs to users and update the service list of the new block.
- **Step 6: Service List Check**. Miners check the service list updated by Step 5 and reject any user request that would exceed the global service list's accommodation limit. Subsequently, transactions representing these requests are marked as rejection, and the rejected users are removed from the service list field.
- **Step 7: Block Generation**. Miners transfer the service fees from the unrejected users to their paired LDs and package all transactions into the request transactions field. Then, the new block is generated and added to the end of the blockchain.
- **Step 8: Notification to Rejected Users**. Based on the request transactions field of the new block, the WiFi AP obtains the rejected users' IDs and notifies them via the WiFi link. These users can resubmit their requests on demand.
- **Step 9: Local Service List Update**. Based on the service list field of the new block, each LD updates its local service list.
- **Step 10: Block Confirmation**. After the new block is generated, several blocks are required to be built on top of it as confirmations to guarantee its tamper resistance.
- **Step 11: Data Transmission**. After the block with the service list is accepted by the main chain, LDs transmit the requested data specified in the transactions to users respectively via secure LiFi downlinks according to the local service lists. Once receiving the requested data, users send ACKs to the WiFi AP via the WiFi link, since the LiFi link is one-way.

During the block confirmation, any other users can also upload requests and ultimately download their requested data securely by repeating Step 3 to Step 11. In the supplementary material, we have shown how the above workflow is implemented via Ethereum Testnet in Appendix E.

III. Case Study

In this section, we provide a specific case study based on Ethereum Testnet to validate the feasibility of the proposed B-HLW architecture and demonstrate its safeguard capability against different threats. Ethereum Testnet is a separate and independent branch of the Ethereum blockchain network, which provides an environment for developing and testing without affecting Mainnet.

We consider a simple scenario where we have a WiFi AP and two LiFi APs, and illustrate the case study in Fig. 4. First, all participants sign the STSC and deploy it on the

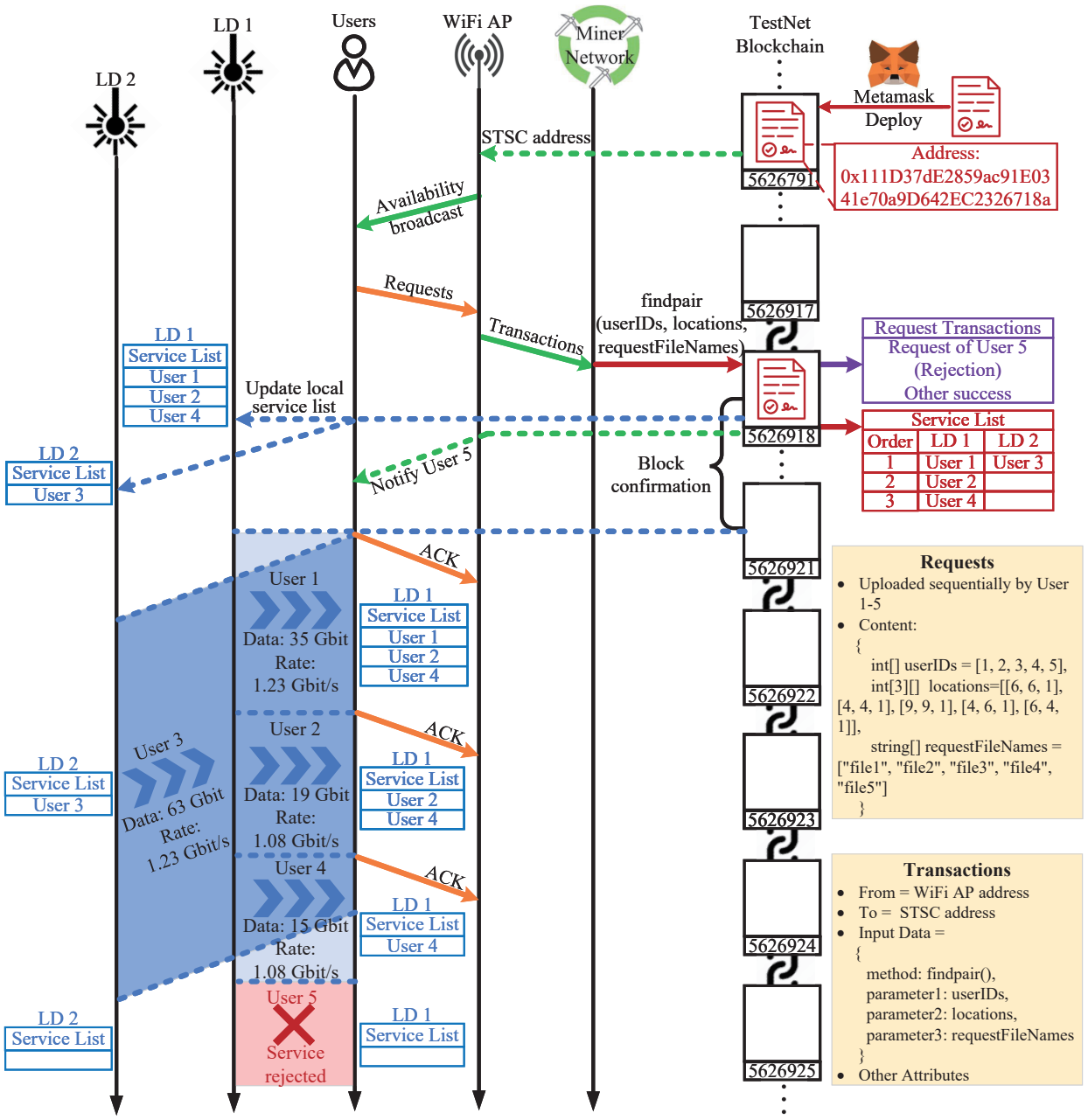


Figure 4: Case study of the B-HLW based on Ethereum. Users' requests is first submitted to the WiFi AP and further committed to Ethereum Testnet through request transactions. Miners include these transactions in a new block (Block 5626918) and execute the pre-deployed STSC for downlink scheduling. As the block gets confirmed, LDs update their local service lists and serve waiting users in sequence.

Testnet blockchain through Metamask. With the STSC's on-chain address, the WiFi AP broadcasts the service availability periodically. Subsequently, a sequence of service requests is submitted by Users 1-5 to the WiFi AP. These requests are then constructed into transactions and committed to Ethereum Testnet by the WiFi AP. The specific contents of both the requests and transactions are shown in Fig. 4. The called STSC automatically processes these requests and records the allocation result on the latest generated block. Since Users 1, 2, 4, and 5 are allocated to LD 1, User 5's request is rejected to avoid exceeding the global service list's accommodation limit. Based on the service list field and request transactions field of this latest generated block, LDs update their local service lists and WiFi AP notifies the rejected user, respectively. Once this block receives sufficient confirmations (4 blocks in Fig. 4) and is accepted into the main chain, LDs transmit the requested data to users in the order of their local service lists, respectively. ACKs are sent by users to the WiFi AP to indicate that they have received the requested data. Note that User 5 is not served by LD 1 as its request has been rejected.

This implementation shows the practical feasibility of B-HLW based on the Ethereum blockchain. The roles of different components of B-HLW are demonstrated throughout this case. For instance, the function `findPair()` workflow effectively assigns the most appropriate LDs to Users 1-5, and the service list recording these allocation results is transparently

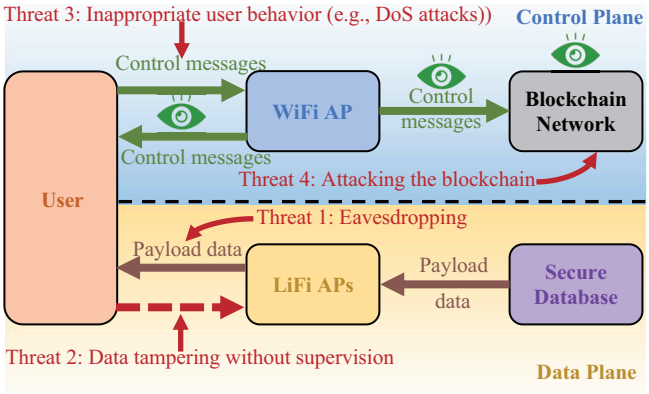


Figure 5: Representative threats to data confidentiality and network supervision and the countermeasures of the B-HLW. The B-HLW separates the data plane and control plane and allows each to handle attacks effectively. In the data plane, the physical property and one-way nature of OWC prevent eavesdropping (Threat 1) and data modification in reverse (Threat 2). In the control plane, various improper user behaviors (Threat 3) and attacks targeting blockchain (Threat 4) can be defended against by the transparent, decentralized, and consensus-based nature of blockchain.

observable on the blockchain. The case study demonstrates that the separation of the control plane and data plane ensures network supervision and data confidentiality simultaneously.

The above case demonstrates the functional workflow. Next, we demonstrate how the B-HLW can address representative threats to data confidentiality and network supervision. Fig. 5 gives several critical attack vectors listed as follows.

1) Eavesdropping: Adversaries can intercept or eavesdrop on downlink transmissions to obtain confidential data.
2) Data Tampering: Malicious users may launch the injection attacks to tamper with confidential data in the database, which would be a more severe form of attack.
3) User Misbehavior: Attackers may launch denial-of-service (DoS) or flooding attacks to obstruct legitimate communications and degrade network performance.
4) Blockchain Attacks: Adversaries might attack the blockchain directly by, e.g., launching an alternative history attack to Block 5626918 in Fig. 4, to alter control records or disrupt network consensus by compromising multiple blockchain nodes.

We design the B-HLW to safeguard against these threats through the coordinated defense of its data plane and control plane. First, eavesdropping is inherently resisted by the physical property of LD-based OWC thanks to its small coverage area and strong directionality [10]. Second, the nature of OWC guarantees the information to be transmitted one-way from LDs to users only, which fundamentally eliminates the possibility of data tampering and any other injection attacks [10]. Third, the transparency of blockchain ensures the visibility and traceability of all control messages [12]. Consequently, misbehavior and malicious activities are accountable and can be detected promptly. Fourth, the decentralized and consensus-based nature of the blockchain ensures that, even if several nodes are attacked, the overall system remains operational and trustworthy [14]. In a nutshell, by defending against the aforementioned threats, the B-HLW achieves data confidentiality in the data plane and network supervision in the control plane.

Moreover, OWC can utilize the spatial transmission potential more effectively. Different LDs can simultaneously use the huge frequency band to serve different users without significant interference (e.g., as shown in Fig. 4, LD 1 serves User 1 and, at the same time, LD 2 serves User 3). Such spatial multiplexing ensures that the B-HLW will not compromise the transmission rate, and conversely, can significantly enhance the downlink transmission rate [10]. The following section will comprehensively illustrate the B-HLW performance in terms of throughput and reliability via simulations.

IV. Simulation Results

We have shown the security threats and illustrated how the B-HLW prevents them in Section III. In this section, we conduct simulations based on the proposed B-HLW architecture from the aspects of throughput, efficiency, and scalability. Due to the page limit, we put the WiFi and LiFi channel models in Appendices A and B in the supplemental material, respectively, and provide detailed simulation configurations in Appendix C.

In Fig. 6(a), we first compare the rejection probability of the B-HLW and WiFi-baselines under varying traffic intensities. The traffic intensity is described by the probability that a user has a request to the database in a unit time. As shown in Fig. 6(a), the B-HLW keeps a lower rejection probability than the WiFi-baselines, particularly under the conditions with more users and high traffic intensities. We can learn from Fig. 6(a) that the B-HLW improves network scalability and reliability via downlink scheduling enabled by our designed blockchain and STSC (see Section II-C). A request will be rejected if too many requests are waiting before it (often caused by a heavy traffic intensity), or if there are no available LDs. We use p_{queue} and p_{LD} to represent the rejection probabilities caused by these two cases, respectively. Fig. 6(a) shows that, as the traffic becomes heavy, the rejection probability increases mainly because user requests are more likely to be denied by the B-HLW to avoid lengthy waiting times.

Fig. 6(b) further compares the throughput of the B-HLW and WiFi-baselines under varying traffic intensities. As shown by Fig. 6(b), thanks to the huge bandwidth and spatial multiplexing of OWC, the B-HLW can achieve the maximum value of 23.7 Gbps and outperforms the WiFi-baselines, indicating that B-HLW can significantly enhance the downlink transmission. Note that the throughput of the WiFi-baselines obtained from the simulation is close to the theoretical maximum throughput of Wi-Fi 6 standard (9.6 Gbps) [13], as well as our theoretical estimation calculated in Appendix D in the supplemental material.

As illustrated in Fig. 6(c) and Fig. 6(d), we compare the performance of the B-HLW and WiFi-baselines under different user numbers. In Fig. 6(c), as the number of users increases, the rejection probability p_{queue} gradually increases. This is because more users lead to heavier traffic, while the maximum length of the service list is limited. Meanwhile, p_{LD} gradually decreases, as more users fall within the LDs' field-of-views

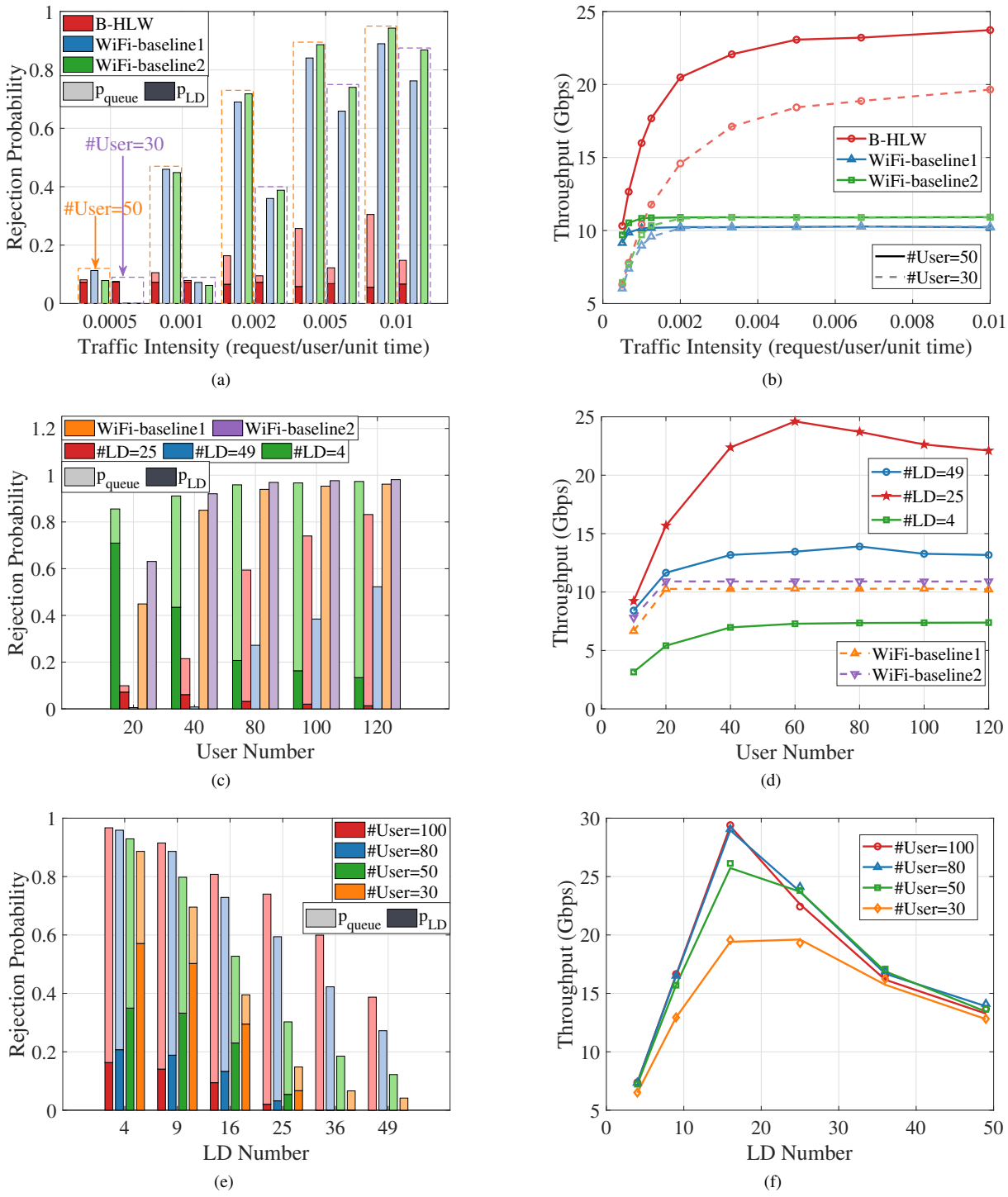


Figure 6: Performance illustration of the B-HLW. (a) Rejection probability with varying traffic intensity (#LD = 25). (b) Throughput with varying traffic intensity (#LD = 25). (c) Rejection probability for different user numbers (traffic intensity = 0.01). (d) Throughput for different user numbers (traffic intensity = 0.01). (e) Impact of LD number on rejection probability (traffic intensity = 0.01). (f) Impact of LD number on throughput (traffic intensity = 0.01).

(FoVs). Despite the overall rejection probability of the B-HLW increasing, it remains lower than that of the WiFi-baselines. Such an advantage becomes more pronounced with more LDs. However, as shown in Fig. 6(d), an excessive number of LDs (#LD = 49) may reduce throughput. Such throughput degradations caused by interference can also be observed when the number of users is large, even with a moderate number of LDs (#LD = 25). From these observations, we can learn the following two lessons. i) Usually, a larger number of LDs implies the broader coverage and enhanced service capability and thus leads to a significantly lower rejection probability than WiFi systems, even in high-load scenarios with a large number of users. It demonstrates the superior scalability of the B-HLW architecture. ii) While a larger number of LDs lowers rejection probability, the growth in user numbers may eventually degrade throughput due to increased interference. Hence, deploying more LDs within a limited indoor space is not always beneficial for system performance.

In Fig. 6(e) and Fig. 6(f), we further illustrate the impact of LD numbers on the performance of the B-HLW. Fig. 6(e) shows that, as the number of LDs increases, both p_{queue} and p_{LD} decrease gradually. p_{LD} even approaches zero when the number of LDs is 49. The reason behind this is that more LDs not only improve service capability but also expand the coverage area. Users are less unlikely to be outside of any LD's FoV. Furthermore, in Fig. 6(f), the throughput initially increases rapidly and then declines as the number of LDs increases. One can learn that, while more users can be served simultaneously by more LDs, the interference among them also becomes more significant. This phenomenon is consistent with our analysis of Fig. 6(d). Therefore, a reasonable configuration of the number of LDs can minimize the rejection probability while providing high-speed transmissions for users.

V. Conclusions

In this paper, we have proposed the B-HLW architecture to resolve the dilemma between data confidentiality and network supervision without sacrificing the transmission rate. We have designed the B-HLW blockchain and smart contract with a complete workflow. We have also demonstrated the feasibility of the B-HLW through a specific Ethereum-based case study and shown what specific threats the B-HLW addressed in this case study to resolve the confidentiality-supervision dilemma. Simulation results have shown that, compared with WiFi-based schemes, the B-HLW provides greater throughput and lower rejection probabilities, which demonstrates its enhanced transmission rates, efficiency, and scalability.

At last, we would like to discuss several practical challenges that should be addressed from theory to real-world deployment.

A. Cost

The cost of B-HLW arises from both LiFi and blockchain. On the LiFi side, the power consumption and potential maintenance expenses impose operating costs. One method to reduce them is to implement intelligent power management, e.g., traffic- and occupancy-aware shutdown. LiFi's installation cost can be lowered by integrating LiFi transceivers into existing lighting fixtures. On the blockchain side, blockchain maintenance requires distributed nodes with computing hardware, and the consensus process consumes significant electrical resources. This challenge can be addressed by adopting energy-efficient consensus protocols, such as Proof of Stake (PoS), which eliminates the need for massive power consumption associated with traditional Proof of Work (PoW) mining.

B. Scalability

As the system scales, blockchain may become a performance bottleneck of B-HLW. The volume of transactions on the blockchain will grow significantly as the number of users and LDs increases. Due to the finite size of individual blocks, certain transactions may not be immediately included in the next available block, which results in a significant increase in access latency. An approach to address this challenge is to simply increase the block size. However, larger blocks require more time to propagate across the network, which increases the risk of temporary forks and weakens the consistency of the ledger. Another more promising solution is to adopt high-throughput blockchain protocols or structures such as Prism [15]. By decoupling transaction propagation from consensus process, Prism can effectively improve throughput without increasing forks and enable the system to approach the physical limits of network capacity.

C. Latency

The inherent confirmation latency of blockchain poses a significant challenge for B-HLW in practical application scenarios with stringent real-time requirements, such as financial transactions and industrial automation. To address this issue, B-HLW can be deployed on a consortium blockchain utilizing the Practical Byzantine Fault Tolerance (PBFT) protocol. Unlike traditional consensus mechanisms, PBFT achieves consensus and confirms block finality through direct communication and voting among a set of pre-authorized nodes. This mechanism avoids network-wide broadcasting and eliminates the need to wait for multiple subsequent block confirmations, thereby reducing the latency from seconds to the millisecond range.

References

[1] V. L. Nguyen, P. C. Lin, B. C. Cheng, R. H. Hwang, and Y. D. Lin, "Security and privacy for 6G: A survey on prospective technologies and challenges," *IEEE Commun. Surv. Tutorials*, vol. 23, no. 4, pp. 2384–2428, Fourthquarter 2021.

[2] W. Lim, K. S. C. Yong, B. T. Lau, and C. C. L. Tan, "Future of generative adversarial networks (GAN) for anomaly detection in network security: A review," *Comput. Secur.*, vol. 139, p. 103733, Apr. 2024.

[3] Y. Liu, Z. Su, H. Peng, Y. Xiang, W. Wang, and R. Li, "Zero trust-based mobile network security architecture," *IEEE Wireless Commun.*, vol. 31, no. 2, pp. 82–88, Apr. 2024.

[4] S. Subramani, S. M, K. A, and S. K. Svn, "Review of security methods based on classical cryptography and quantum cryptography," *Cybern. Syst.*, vol. 56, no. 3, pp. 302–320, Jan. 2025.

[5] H. Niu, Y. Xiao, X. Lei, J. Chen, Z. Xiao, M. Li, and C. Yuen, "A survey on artificial noise for physical layer security: Opportunities, technologies, guidelines, advances, and trends," *IEEE Commun. Surv. Tutorials*, vol. 28, pp. 341–381, Sep. 2026.

[6] J. Chu, R. Liu, M. Li, Y. Liu, and Q. Liu, "Joint secure transmit beamforming designs for integrated sensing and communication systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 4, pp. 4778–4791, Apr. 2023.

[7] X. Ling, P. Cheng, Y. Le, J. Wang, and X. You, "Blockchain-based network hawkeye function: Building the trust triangle in 6G," *IEEE Network*, vol. 36, no. 4, pp. 104–111, Aug. 2022.
[8] M. Keshk, E. Sitnikova, N. Moustafa, J. Hu, and I. Khalil, "An integrated framework for privacy-preserving based anomaly detection for cyber-physical systems," *IEEE Trans. Sustainable Comput.*, vol. 6, no. 1, pp. 66–79, Mar. 2021.
[9] S. Gyawali, Y. Qian, and R. Q. Hu, "A privacy-preserving misbehavior detection system in vehicular communication networks," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 6147–6158, Jun. 2021.
[10] A. Celik, I. Romdhane, G. Kaddoum, and A. M. Eltawil, "A top-down survey on optical wireless communications for the internet of things," *IEEE Commun. Surv. Tutorials*, vol. 25, no. 1, pp. 1–45, Firstquarter 2023.
[11] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," *Tech. Rep.*, Oct. 2008.
[12] X. Ling, Y. Le, S. Chen, J. Wang, and X. Zhou, "Blockchain-enabled decentralized services and networks: Assessing roles and impacts," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 6, pp. 2141–2154, Jun. 2025.
[13] E. Khorov, A. Kiryanov, A. Lyakhov, and G. Bianchi, "A tutorial on IEEE 802.11ax high efficiency WLANs," *IEEE Commun. Surv. Tutorials*, vol. 21, no. 1, pp. 197–216, Firstquarter 2019.
[14] X. Ling, Y. Le, J. Wang, Y. Huang, and X. You, "Trust and trustworthiness in information and communications technologies," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 84–92, Apr. 2025.
[15] V. Bagaria, S. Kannan, D. Tse, G. Fanti, and P. Viswanath, "Prism: Deconstructing the blockchain to approach physical limits," in *Proc. 26th ACM SIGSAC Conf. Comput. Commun. Secur. (CCS'19)*, London, UK, Nov. 2019, pp. 585–602.